# Effects of dietary protein on composition and metabolism of plasma lipoproteins in rabbits.

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Changes in the concentration and composition of serum VLDL, LDL, and HDL were studied in rabbits transferred from Chow diets to cholesterol-free, semipurified diets containing casein or isolated soy protein. During the first week on the casein diet, there was a marked increase in LDL-cholesterol and these higher levels were maintained during the subsequent 3 weeks of the study. Similar but less marked changes were obtained with the soy protein diet. When the percent composition of the particles was determined, both VLDL and LDL had a higher proportion of cholesterol. Turnover studies indicated that the FCRs for radiolabelled VLDL and LDL were reduced in casein-fed animals compared to those fed soy protein. The elevated LDL levels in casein-fed rabbits were primarily due to a reduction in receptor-mediated catabolism of LDL-apo B. Receptor-independent removal in the two groups was similar. These studies show that the hypercholesterolemia in casein-fed rabbits, compared to those fed soy protein, is associated with cholesterol enrichment of LDL and impaired receptor-dependent removal of LDL-apo B.